

Cetuximab preclinical antitumor activity (monotherapy and combination based) is not predicted by relative total or activated epidermal growth facto...



Although Erbitux (cetuximab) has proven therapeutic benefit in the clinical setting, the molecular determinants predicting responsiveness to this agent are still not very well understood. Here, we assessed the relationship between basal total and activated (pY1068) epidermal growth factor receptor (EGFR) levels in a tumor and the responsiveness to cetuximab monotherapy or combination-based treatment using human xenograft models. Cetuximab treatment alone (0.25-1 mg/mouse/injection, q3d, i.p.) effectively delayed the growth of GEO and L2987 tumors by a minimum of 10 days corresponding to log cell kill values of ≥ 1.0 . Borderline activity was seen in the A549 and WiDr xenografts. However, cetuximab failed to show any significant antitumor activity in the HT29, HCT116, LOVO, Colo205, LX-1, HCC70, and N87 models. All of the studied tumors had detectable yet variable levels of EGFR. For combination regimens, cetuximab (1 mg/mouse/injection, q3dx5, i.p.) and cisplatin (4.5 mg/kg/injection, q3dx5, i.v.) proved to be significantly more efficacious than individual monotherapies in the cisplatin-refractory yet cetuximab-responsive GEO tumor model ($P < 0.001$). However, no therapeutic enhancement was observed in the cisplatin and cetuximab weakly responsive A549 xenograft. Similarly, combinations of CPT-11 (48 mg/kg/injection, q3dx5, i.v.) with cetuximab (1 mg/mouse/injection, q3dx5, i.p.) failed to show any improvements over individual monotherapies in the cetuximab resistant/weakly responsive HT29, A549, and WiDr models. We conclude that preclinical activity associated with cetuximab monotherapy does not correlate directly with relative basal levels of total or activated (pY1068) EGFR in a tumor. Moreover, robust single-agent activity by cetuximab may be the best predictor for this agent to potentiate chemotherapy-mediated antitumor activities.